

Gerald Emanuel Vargas Morales

Computer Engineering Student · Full Stack Developer · Data Analysis · Applied AI

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PROFILE

Final-year Computer Engineering student at **Universidad Mayor de San Simón (UMSS)** with solid technical training and hands-on experience in high-criticality projects. Contributed to the development of **electoral systems** for the Cochabamba Sub-national Elections, mobile applications for the Municipal Government of Colcapirhua, and modern educational platforms. Committed to technical excellence and continuous learning, with a strong focus on **Artificial Intelligence** and **Machine Learning**.

PROFESSIONAL EXPERIENCE

Full Stack Developer — Official Internship

Gobierno Autónomo Municipal de Colcapirhua · Laravel 11 + Flutter 3.41

January 2026 – Present

- ▷ Developed **ColcaTrufis**: cross-platform urban mobility app (Trufi/Radiotaxi) with official municipal route data.
- ▷ Stack: **Flutter** (mobile frontend) + **Laravel** (RESTful APIs). Development and maintenance of institutional websites.
- ▷ Digital solutions for optimising municipal administrative processes.

🔗 View Live Site

📄 Download APK

Full Stack Developer — Electoral Ballot System (NGP Party)

Sub-national Elections Cochabamba · Node.js + React

December 2025 – March 2026

CURRICULAR CERTIFICATE

- ▷ **Successfully deployed** and used during the official Cochabamba sub-national electoral process.
- ▷ Backend: **Node.js/Express** — RESTful APIs for real-time electoral data processing.
- ▷ Frontend: **React** — interactive dashboard for live visualisation of results. Scalable clean architecture.

📄 Download Electoral App

Full Stack Mobile Developer — Electricity Meter Reading System

Cooperativa de Electricidad “15 de Noviembre” — Potosí, Bolivia · Flutter · Laravel · ESC/POS · Bluetooth · OpenStreetMap

FIRST PAID JOB Android mobile application developed for field technicians of the “15 de Noviembre” electrical cooperative (Potosí, Bolivia). Enabled on-site meter reading registration with GPS geolocation and automatic thermal receipt printing via Bluetooth. Backend built with **Laravel**, consuming the cooperative’s existing system REST APIs. Full management of Bluetooth peripheral lifecycle and runtime Android permissions.

- ▷ Field technician authentication and meter reading registration with GPS coordinates.
- ▷ Automatic pre-notice printing via **Bluetooth ESC/POS** thermal printer upon each reading.
- ▷ Full thermal printer management: device scanning, pairing, and automatic reconnection.
- ▷ Interactive client map rendered with **flutter_map** + **OpenStreetMap** (no API key required).
- ▷ REST API integration with the cooperative’s legacy system; local printer config persistence.

Process Automation with n8n

Automation Projects (in collaboration)

March 2026

- ▷ Automated bulk migration of **6,500+** **Excel rows** into the ballot system with full traceability.
- ▷ Multi-source data integration and validation via **n8n** workflow orchestration.

FEATURED PROJECTS

Synapse — Comprehensive Educational Platform

Laravel 12 · PHP 8 · MySQL · JavaScript

2025 – 2026

- ▷ Full-featured LMS: course management, user administration and academic content delivery.

- ▷ Role-based access (Student / Teacher / Admin) with personalised dashboards per profile.
- ▷ Automated assessments, progress tracking, and detailed admin reports. Optimised with caching and eager loading.

Financial Fraud Detection with Deep Learning

Python · Streamlit · TensorFlow/Keras · Scikit-learn · Pandas · Plotly

2025 – 2026

- ▷ Hybrid **Isolation Forest** + **Autoencoder** system (20-12-8-12-20 architecture) for anomaly detection.
- ▷ Pipeline for imbalanced data (fraud <0.1%): **94.2% precision** / **91.8% recall** on 150,000+ transactions.
- ▷ Interactive Streamlit dashboard with anomaly visualisation and adaptive risk scoring.

TECHNICAL SKILLS

Backend	Python, PHP, Laravel, Node.js, Express.js, Java, C++
Frontend	React, Flutter, HTML5, CSS3, JavaScript
Data / ML	Pandas, NumPy, TensorFlow, Keras, Scikit-learn, Streamlit
Databases	MySQL, PostgreSQL, MongoDB, SQLite, SQL Server
Tools	Git, GitHub, n8n, Docker, Linux, Jira

EDUCATION

Bachelor of Computer Engineering

Universidad Mayor de San Simón (UMSS) — Cochabamba, Bolivia

2024 – Present

CERTIFICATIONS & COURSES

- ▷ Intensive Python (NumPy, Pandas)
 - ▷ Machine Learning (Microsoft)
 - ▷ Recursion and Backtracking
- ▷ Competitive Programming — ICPC-UMSS
 - ▷ Web Development with Laravel

ACHIEVEMENTS

Curricular Certificate: Official Electoral Ballot System — Sub-national Elections Cochabamba 2026.

ICPC Cochabamba 2024: Ranked **#17** in first competitive team participation.

Production Deployment: ColcaTrufis live at moviruta.colcapirhua.gob.bo.

Successful Electoral System: Actively used in the official Cochabamba 2026 electoral process.

LANGUAGES

Spanish Native

English B1
